Credit Name: CSE 2140 2nd Language Programming

Assignment Name: digits.java

**The following questions can help you in thinking critically about your problem-solving processes:**

**Understanding the Problem**

How did you approach understanding the challenge?

I started by reading the excercise prompt to figure out what outputs were required (hundreds, tens, and ones digits). Then i looked at the textbook example and sample output to understand how the 3 digit number should be broken down

Were there any parts of the problem you found confusing at first? If so, how did you resolve that confusion?

Understanding how to extract the middle (tens) digits using the math operators was a bit tricky at first. I resolved this by getting feedback from my desk mate and my teacher

**Planning the Solution**

Did you create a plan or break the problem into smaller steps before coding?

I broke it down into four main steps. 1. taking care of the user input with Scanner 2. Calculating the hundreds digit 3. calculating the tens digit 4. calculating the ones digit

How did you decide on the tools, data structures, or algorithms to use?

I chose int variable because digits and inputs are whole numbers, used integer division / to drop remainders for higher place values and modulus % to isolate the remaining digits i had

**Implementation**

Did you write the code in small pieces or attempt the entire solution at once?

i wrote the code in small pieces , first i set up the scanner and took user input, and then added the formulars for each digit place one by one

How did you test your solution along the way to make sure it was working?

Not the entire way, but nearing the end i tested the program using sample inputs like 256 to check if the outputs for hundreds, tens, and ones matched the textbooks expected output

**Overcoming Challenges**

What part of the problem was the most difficult for you?

The most difficult part was the logic for the tens place calculations ((digits / % 100) / 10) as it required using both % and / operations

How did you handle moments when you felt stuck or unsure of what to do next?

I double checked the formulas or asked my deskmate

**Learning**

Was there anything you learned that you think will help you with future challenges?

I learned how useful integer division and the modulus operator are for isolating specific digits from a larger number, and it could definetly help me with future math based excercises